

Scale free property and edge state of Wilson's numerical renormalization group

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We discuss the scale-free property of Wilson's numerical renormalization group(NRG) for the Kondo impurity problem. The single-particle state of the effective Hamiltonian with a cutoff Λ is described by the wavepacket basis having the scale free property; The energy scale of the system can be controlled by the lattice translation of the wavepacket basis with no reference of rescaling of the lattice space. We also analyze the role of the Kondo interaction in the context of wavepacket basis and then discuss the scaling and renormalization of the Kondo coupling. In addition, we clarify the role of the edge state in the lowest energy scale of Wilson NRG.

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I. INTRODUCTION

Renormalization group(RG) has been one of the most fundamental concept in statistical physics to explain hierarchical structure of the energy scale in nature.¹ In particular, such a real-space RG approach as block-spin transformation provides an intuitive view for coarse graining of the fluctuation field and rescaling of the length unit; It is well-known that the block-spin transformation for the triangular lattice Ising model is often discussed as a textbook example². For the low-dimensional quantum RG, however, the situation is different; The conventional block-spin transformation often fails in extracting the low-energy physics of the one-dimensional(1D) quantum system, and improvement of the real-space RG has been an attractive frontier in the quantum physics. In modern view, this is mainly because the block-spin transformation breaks the quantum entanglement associated with the large quantum fluctuation.^{3,4}

Historically, two real-space RG approaches not based on the block-spin transformation made great success in the 1D quantum physics: Wilson's numerical renormalization group(NRG)⁵⁻⁷ for the Kondo impurity problem⁸ and density matrix renormalization group(DMRG)^{9,10}. An interesting point is that these two methods have completely different characters. Wilson NRG can directly deal with the metallic system with a magnetic impurity, through mapping to the effective 1D chain with a boundary. On the other hand, we should remark that DMRG includes no scale transformation and no concept of the running coupling constant. This implies that DMRG should be considered as a variational optimization method for the matrix product wavefunction¹¹⁻¹⁵, rather than a RG based on the coarse graining and the scale transformation. Thus, Wilson's pioneering work has been the peculiar RG method that succeeds in such energy-scale control of the massless excitation. What is its key mechanism? Answer to this question provides an essential insight for the quantum RG. Here, we just note that, recently, the entanglement renormalization interestingly reproduces the correct critical index for 1D quantum systems¹⁶.

In this paper, we will show that Wilson NRG has the scale free property, which is the key of systematical selectivity of the arbitrary energy scale around the Fermi surface. For this purpose, we directly focus on the relation between the wavefunction and the cutoff parameter Λ due to the log-discretization. In actual NRG iterations, a free fermion site is added step by step with decreasing the energy scale by Λ and the effective 1D tight-binding model having the exponential modulation of the hopping parameter is eventually generated. The qualitative role of this cutoff can be viewed as a regulator for the infrared divergence, which introduces effective length scale $\xi \sim 1/\ln \Lambda$.¹⁷ In actual numerical computations, however, the cutoff value is usually adopted to be $\Lambda = 1.5 \sim 2.5$, which is beyond perturbation level to the original Kondo problem($\Lambda = 1$). The success of Wilson NRG with the nonperturbative value of Λ suggests that the Wilson's cutoff scheme should involve some plausible mechanism to maintain the nature at $\Lambda = 1$; We clarify that the one-particle eigen-wavefunction is modified from the plane wave into the wavepacket by the cutoff, and its lattice translation explains the selectivity of the energy scale. We also find that there is an edge state in the lowest energy scale near the Fermi surface. On the basis of the wavepacket basis above, we perform the exact diagonalization of the Wilson effective Hamiltonian including the Kondo impurity. We then discuss that the scale factor of the groundstate wavefunction should be renormalized from the value of the non-interacting wavepacket by the Kondo interaction.

This paper is organized as follows. In section II, we define the effective tight-binding model and then explain its scale free property. In section III, nature of the wavepacket is analyzed. In section IV, we resolve the role of the Kondo iteration in the context of the wavepacket basis. Section V is devoted to the summary and discussion.